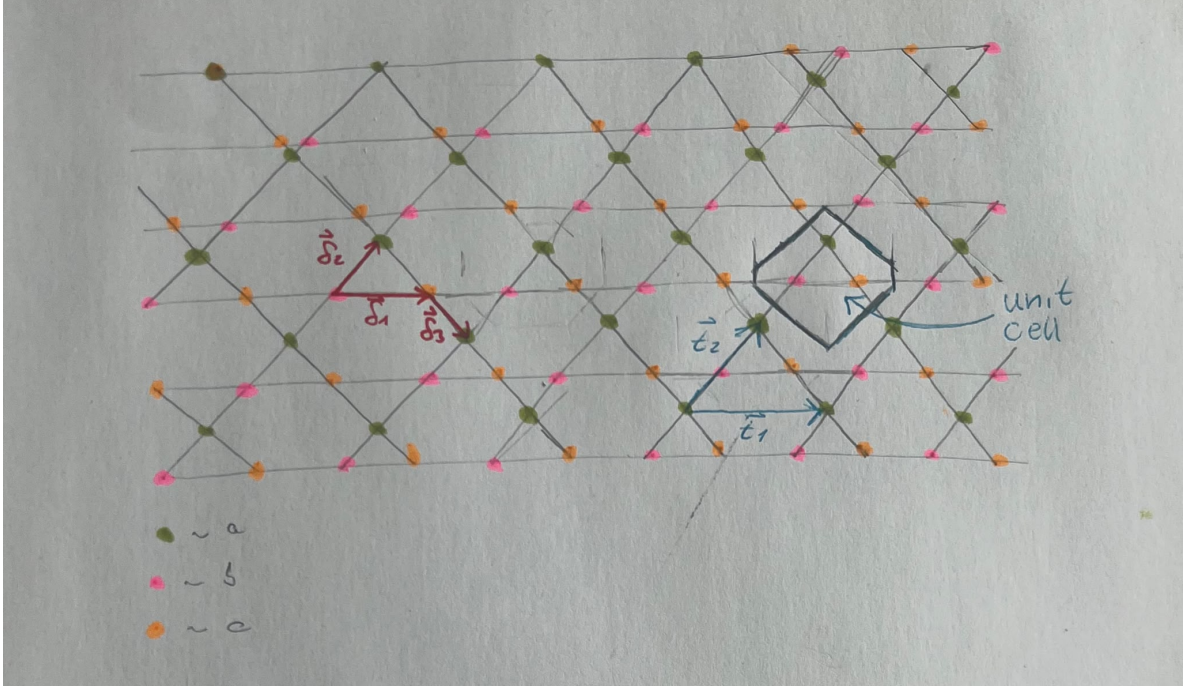


0.1 0-flux State

0.1.1 Real Space

The bare Kagome lattice is a triangular Bravais lattice with a three site unit cell (called a, b and c-sites below). The Wigner Seitz-cell is a hexagon (with full symmetries, even if the drawing suggests otherwise ;)). Geometrically speaking the lattice consists of corner-sharing equilateral triangles.



In all that follows I normalize all distances wrt the distance between nearest neighbours. The nearest neighbour vectors are (see cartoon)

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} \quad (1)$$

and the primitive translations

$$\vec{t}_1 = 2\vec{\delta}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = 2\vec{\delta}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}. \quad (2)$$

The tight binding hamiltonian is

$$H = t \sum_{i\sigma} \left(a_{i\sigma}^\dagger c_{i\sigma} + c_{i\sigma}^\dagger b_{i\sigma} + b_{i\sigma}^\dagger a_{i\sigma} + c_{i\sigma}^\dagger b_{i+t_1;\sigma} + c_{i\sigma}^\dagger a_{i+t_1-t_2;\sigma} + a_{i\sigma}^\dagger b_{i+t_2;\sigma} + \text{h.c.} \right) \quad (3)$$

The first index refers to the unit cell and the second one to the "species" of the particle, the latter being a product of the parton decomposition.

0.1.2 Reciprocal Space

The reciprocal lattice vectors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix} \quad (4)$$

The first Brillouin zone is also a hexagon (rotated by 90° wrt to the Wigner-Seitz cell). To find H in reciprocal space I do a Fourier transform:

$$a_{i\sigma} = \sqrt{\frac{3}{N}} \sum_k e^{i\vec{k}\cdot\vec{R}_i} a_{k\sigma}, \quad (5)$$

where N is the number of unit cells. This yields

$$H = 2t \sum_{q\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger & c_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 0 & \cos(\vec{q} \cdot \vec{\delta}_2) & \cos(\vec{q} \cdot \vec{\delta}_3) \\ \cos(\vec{q} \cdot \vec{\delta}_2) & 0 & \cos(\vec{q} \cdot \vec{\delta}_1) \\ \cos(\vec{q} \cdot \vec{\delta}_3) & \cos(\vec{q} \cdot \vec{\delta}_1) & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix}. \quad (6)$$

0.1.3 dispersion and band touching points

The three eigenvalues are

$$\varepsilon_1(\vec{q}) = t \left(1 - \sqrt{4 \cos^2(q_x) + 4 \cos(q_x) \cos(\sqrt{3}q_y) + 1} \right) \quad (7)$$

$$\varepsilon_2(\vec{q}) = t \left(1 + \sqrt{4 \cos^2(q_x) + 4 \cos(q_x) \cos(\sqrt{3}q_y) + 1} \right) \quad (8)$$

$$\varepsilon_3(\vec{q}) = -2t, \quad (9)$$

see [1]. I also used

$$\cos^2\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) + \cos^2\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) = \cos(q_x) \cos(\sqrt{3}q_y) + 1 \quad (10)$$

Interestingly, the top band is completely flat. The other two bands touch when

$$\cos^2(q_x) + \cos(q_x) \cos(\sqrt{3}q_y) + 1/4 = 0 \quad (11)$$

$$\Rightarrow \cos(q_x) = \frac{1}{2} \left(-\cos(\sqrt{3}q_y) \pm \sqrt{\cos^2(\sqrt{3}q_y) - 1} \right) \quad (12)$$

$$\Rightarrow q_y = \frac{\pi l}{\sqrt{3}}, \quad l \in \mathbb{N} \quad (13)$$

$$\Rightarrow q_x = (-1)^s \left(\frac{\pi}{2} + \frac{(-1)^l \pi}{6} \right) + 2\pi m, \quad m \in \mathbb{N}, \quad s \in [0, 1] \quad (14)$$

A handy solution to this equation is

$$\begin{aligned} q_x &= (-1)^s \frac{2\pi}{3} + m_1 \pi \\ q_y &= \frac{\pi}{\sqrt{3}} (2m_2 - m_1), \end{aligned} \quad (15)$$

with $s \in 0, 1$ and $m_1, m_2 \in \mathbb{N}$. These touching points lie at the corners of the Brillouin zone, they come in two flavours. The parameter s labels the aforementioned flavours while m_1 (m_2) is a translation by the reciprocal lattice vector $\vec{b}_1$ ($\vec{b}_2$). For example

$$\vec{K} = \begin{pmatrix} 2\pi/3 \\ 0 \end{pmatrix}, \quad s = 0, \quad m_1 = 0, \quad m_2 = 0 \quad (16)$$

$$\vec{K}' = \begin{pmatrix} -2\pi/3 \\ 0 \end{pmatrix}, \quad s = 1, \quad m_1 = 0, \quad m_2 = 0. \quad (17)$$

0.1.4 Filling Factor

The idea is the following: The lower two bands contain the same number of states by symmetry. By calculating the number of states of the upper band (should be easy) it should be possible to calculate at what filling factor the low energy physics happen exactly the linear touching points.

0.1.5 Linearization in Reciprocal Space and Projection

Use

$$\cos\left(\left(\vec{K}^{(\prime)} + \vec{p}\right) \cdot \vec{\delta}_{ij}\right) \approx \cos\left(\vec{K}^{(\prime)} \cdot \vec{\delta}_{ij}\right) - \sin\left(\vec{K} \cdot \vec{\delta}_{ij}\right) \left(\vec{p} \cdot \vec{\delta}_{ij}\right) \quad (18)$$

and

$$\begin{aligned} \vec{K} \cdot \vec{\delta}_1 &= \frac{2\pi}{3} \\ \vec{K} \cdot \vec{\delta}_2 &= \frac{\pi}{3} \\ \vec{K} \cdot \vec{\delta}_3 &= \frac{\pi}{3} \\ \vec{K}' &= -\vec{K} \end{aligned} \quad (19)$$

$$H_{\vec{K}} \approx t \sum_{p\sigma} \begin{pmatrix} a_{p\sigma}^\dagger & b_{p\sigma}^\dagger & c_{p\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 0 & 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_3 \\ 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 0 & -1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_1 \\ 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_3 & -1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_1 & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix} \quad (20)$$

Directly at the $\vec{K}$ -point the eigenvalues are obviously

$$\varepsilon_1(\vec{K}) = t, \quad \varepsilon_2(\vec{K}) = t, \quad \varepsilon_3(\vec{K}) = -2t, \quad (21)$$

the eigenvector corresponding to the flat band reads at the $\vec{K}$ -point:

$$\vec{v}_3(\vec{K}) = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}. \quad (22)$$

Since H is hermitian (/symmetric in this case) there exist a orthonormal basis of eigenvectors. Obviously the subspace orthonormal to $\vec{v}_3(\vec{K})$ corresponds to the two bands that cross. A possible (orthonormal) basis for this subspace is

$$\vec{v}_1(\vec{K}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{v}_2(\vec{K}) = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}. \quad (23)$$

Projecting H onto this basis yields

$$\begin{aligned} \vec{v}_1^T \cdot H_{\vec{K}} \cdot \vec{v}_1 &= t - \sqrt{3}t\vec{\delta}_2 \cdot \vec{p} & \vec{v}_1^T \cdot H_{\vec{K}} \cdot \vec{v}_2 &= -t(\vec{\delta}_1 + \vec{\delta}_3) \cdot \vec{p} \\ \vec{v}_2^T \cdot H_{\vec{K}} \cdot \vec{v}_1 &= -t(\vec{\delta}_1 + \vec{\delta}_3) \cdot \vec{p} & \vec{v}_2^T \cdot H_{\vec{K}} \cdot \vec{v}_2 &= t + \sqrt{3}t\vec{\delta}_2 \cdot \vec{p} \\ & \Leftrightarrow t\sigma_0 - \sqrt{3}t(\vec{\delta}_2 \cdot \vec{p})\sigma_z + t((\vec{\delta}_1 + \vec{\delta}_3) \cdot \vec{p})\sigma_x \end{aligned} \quad (24)$$

This already has the correct dispersion

$$\varepsilon_{1,2}(\vec{K} + \vec{p}) \approx \sqrt{3}t|\vec{p}|, \quad (25)$$

but is not yet in the form of the Weyl equation. To transform the equation into the Weyl equation one can rotate the basis inside the degenerate subspace. I take the following approach; define

$$\begin{aligned}\vec{w}_1 &= \alpha \vec{v}_1 + \beta \vec{v}_2 \\ \vec{w}_2 &= \eta \vec{v}_1 + \gamma \vec{v}_2\end{aligned}\tag{26}$$

Othogonality dictates

$$\begin{aligned}|\alpha|^2 + |\beta|^2 &= 1 \\ |\eta|^2 + |\gamma|^2 &= 1 \\ \alpha^* \eta + \beta^* \gamma &= 0.\end{aligned}$$

Maybe make an Ansatz using trig functions and phases? Though I should count free parameters before. Educated guess with 3 free parameters:

$$\begin{aligned}\alpha &= e^{i\theta_1} \sin(\phi) \\ \beta &= e^{i\theta_2} \cos(\phi) \\ \eta &= e^{i\theta_1} \cos(\phi) \\ \gamma &= -e^{i\theta_2} \sin(\phi).\end{aligned}$$

Utilizing $\vec{\delta}_2 + \vec{\delta}_3 = \vec{\delta}_1$ I get:

$$\vec{w}_1^H \cdot H_{\vec{K}} \cdot \vec{w}_1 = t + \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(1 - 2|\alpha|^2 - \frac{2\text{Re}(\alpha^* \beta)}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(-\frac{4\text{Re}(\alpha^* \beta)}{\sqrt{3}} \right) \tag{27}$$

$$\vec{w}_2^H \cdot H_{\vec{K}} \cdot \vec{w}_2 = 1 + \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(1 - 2|\eta|^2 - \frac{2\text{Re}(\eta^* \gamma)}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(-\frac{4\text{Re}(\eta^* \gamma)}{\sqrt{3}} \right) \tag{28}$$

$$\vec{w}_1^H \cdot H_{\vec{K}} \cdot \vec{w}_2 = \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(-2\alpha^* \eta - \frac{\alpha^* \gamma + \beta^* \eta}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(-\frac{2(\alpha^* \gamma + \beta^* \eta)}{\sqrt{3}} \right) \tag{29}$$

$$\vec{w}_2^H \cdot H_{\vec{K}} \cdot \vec{w}_1 = (\vec{w}_1^H \cdot H_{\vec{K}} \cdot \vec{w}_2)^* \tag{30}$$

The strategy is to solve for values for α, β, γ and η such that the following combinations occur in the matrix elements:

$$p_x = (\vec{\delta}_2 + \vec{\delta}_3) \cdot \vec{p} \tag{31}$$

$$p_y = \frac{1}{\sqrt{3}} (\vec{\delta}_2 - \vec{\delta}_3) \cdot \vec{p} \tag{32}$$

whil still leading to orthonormal basis vectors. The following parameters are a sensible choice

$$\alpha = -\frac{1}{2}, \quad \beta = \frac{\sqrt{3}}{2}, \quad \eta = \frac{\sqrt{3}}{2}, \quad \gamma = \frac{1}{2}. \tag{33}$$

Thus, the new basis vectors read

$$\vec{w}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \quad \vec{w}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \tag{34}$$

$$\Rightarrow \langle H_{\vec{K}} \rangle_{\vec{w}_1, \vec{w}_2} = t\sigma_0 + \sqrt{3}tp_y\sigma_x + \sqrt{3}tp_x\sigma_z \tag{35}$$

This form was also found in [2], a different one was obtained in [3]. I expect that I would find the latter if I chose different values for α, β, γ and η . I decided to stick with this form to keep the expressions real.

I could eliminate the σ_0 term by introducing chemical potentials but I guess for non-interacting particles it doesn't really matter.

Question: when rotating the orthonormal basis inside the degenerate subspace there is one free parameter (the rotation angle). On the other hand one can probably also think about the surface of a Bloch sphere (the radius is obviously fixed). What is the free parameter in this picture /what is the fixed parameter? It is probably worthwhile to explore this at some point, it might reveal the underlying topology.

Addition: the counting of free coefficients might not be correct, considering they can be complex. Is the Bloch sphere argument even valid?

0.1.6 Introduction of Strain

I show how to expand expressions prototypical for the k-space Hamiltonian. First I expand in strain:

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{q}\cdot\vec{\delta}_{ij}} \left(1 - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\vec{q} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \right), \quad (36)$$

afterwards I expand around $\vec{K}$:

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{q}\cdot\vec{\delta}_{ij}} \left(1 + \vec{p} \cdot \vec{\delta}_{ij} \right) \left(1 - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\underline{u} \cdot \vec{\delta}_{ij} \right) \cdot \vec{p} \right). \quad (37)$$

This compares well with the field theory method for the triangular lattice where I did the expansions in reverse order.

Keeping only the relevant terms (relevant to this order at least):

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{K}\cdot\vec{\delta}_{ij}} \left(1 + i\vec{p} \cdot \vec{\delta}_{ij} - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) + i \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \right) \quad (38)$$

$$\begin{aligned} \Rightarrow t(-\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} + t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} &\approx 2t \cos(\vec{K} \cdot \vec{\delta}_{ij}) - 2t \sin(\vec{K} \cdot \vec{\delta}_{ij}) \left(\vec{p} \cdot \vec{\delta}_{ij} \right) \\ &\quad - 2\beta t \cos(\vec{K} \cdot \vec{\delta}_{ij}) \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \\ &\quad - 2t \sin(\vec{K} \cdot \vec{\delta}_{ij}) \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \end{aligned} \quad (39)$$

I group the linear corrections by order in β to find the two matrices:

$$\delta H_1 = -2t\beta \begin{pmatrix} 0 & (u_{xx} + 2\sqrt{3}u_{yx} + 3u_{yy})/8 & (u_{xx} - 2\sqrt{3}u_{yx} + 3u_{yy})/8 \\ (u_{xx} + 2\sqrt{3}u_{yx} + 3u_{yy})/8 & 0 & -u_{xx}/2 \\ (u_{xx} - 2\sqrt{3}u_{yx} + 3u_{yy})/8 & -u_{xx}/2 & 0 \end{pmatrix} \quad (40)$$

$$\delta H_2 = -2t \begin{pmatrix} 0 & \pi(\frac{u_{xx}}{\sqrt{3}} + u_{yx})/2 & \pi(\frac{u_{xx}}{\sqrt{3}} - u_{yx})/2 \\ \pi(\frac{u_{xx}}{\sqrt{3}} + u_{yx})/2 & 0 & \pi\frac{u_{xx}}{\sqrt{3}} \\ \pi(\frac{u_{xx}}{\sqrt{3}} - u_{yx})/2 & \pi\frac{u_{xx}}{\sqrt{3}} & 0 \end{pmatrix} \quad (41)$$

The projections are

$$\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2} = -t\beta \begin{pmatrix} u_{xx} & u_{yx} \\ u_{yx} & u_{yy} \end{pmatrix} \quad (42)$$

$$\langle \delta H_2 \rangle_{\vec{w}_1, \vec{w}_2} = -\frac{2\pi t}{\sqrt{3}} \begin{pmatrix} u_{xx} & u_{yx} \\ u_{yx} & -u_{xx} \end{pmatrix} = -\frac{2\pi t}{\sqrt{3}} (u_{yx}\sigma_y + u_{xx}\sigma_z) \quad (43)$$

$\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2}$ can be written as vector potential + scalar potential using:

$$\begin{aligned} u_{xx} &= \frac{1}{2} (u_{xx} - u_{yy}) + \frac{1}{2} (u_{xx} + u_{yy}) \\ u_{yy} &= -\frac{1}{2} (u_{xx} - u_{yy}) + \frac{1}{2} (u_{xx} + u_{yy}). \end{aligned} \quad (44)$$

$$\Rightarrow \langle H_{\vec{K}} + \delta H_1 + \delta H_2 \rangle_{\vec{w}_1, \vec{w}_2} = (t + \phi)\sigma^0 + \sqrt{3}t \left(p_y + A_{\vec{K}}^y \right) \sigma_x + \sqrt{3}t \left(p_x + A_{\vec{K}}^x \right) \sigma_z \quad (45)$$

$$\phi = \frac{t}{2} (u_{xx} + u_{yy}) \quad (46)$$

$$A_{\vec{K}}^x = -\frac{t\beta}{2\sqrt{3}} (u_{xx} - u_{yy}) - \frac{2\pi t}{3} u_{xx} \quad (47)$$

$$A_{\vec{K}}^y = -\frac{t\beta}{\sqrt{3}} u_{yx} - \frac{2\pi t}{3} u_{yx} \quad (48)$$

The β -independent part of $\vec{A}_{\vec{K}}$ can be eliminated using a gauge-transformation. This agrees with the results here [3]. **It bothers me that only u_{yx} appears but not u_{xy} . Check this**

I give a few intermediate results for the derivation of $\vec{A}_{\vec{K}'}$:

The cool kids introduce a cone flavour index and derive the expressions for both flavours at the same time

$$H_{\vec{K}'} \approx t \sum_{p\sigma} \begin{pmatrix} a_{p\sigma}^\dagger & b_{p\sigma}^\dagger & c_{p\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 0 & 1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_3 \\ 1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 0 & -1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_1 \\ 1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_3 & -1 + \sqrt{3}\vec{p} \cdot \vec{\delta}_1 & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix} \quad (49)$$

(the zeroth order is the same so I make the same Ansatz for the initial basis.)

$$\begin{aligned} \vec{v}_1^T \cdot H_{\vec{K}'} \cdot \vec{v}_1 &= t + \sqrt{3}t \vec{\delta}_2 \cdot \vec{p} & \vec{v}_1^T \cdot H_{\vec{K}'} \cdot \vec{v}_2 &= t \left(\vec{\delta}_1 + \vec{\delta}_3 \right) \cdot \vec{p} \\ \vec{v}_2^T \cdot H_{\vec{K}'} \cdot \vec{v}_1 &= t \left(\vec{\delta}_1 + \vec{\delta}_3 \right) \cdot \vec{p} & \vec{v}_2^T \cdot H_{\vec{K}'} \cdot \vec{v}_2 &= t - \sqrt{3}t \vec{\delta}_2 \cdot \vec{p} \end{aligned}$$

$$\vec{w}_1^H \cdot H_{\vec{K}'} \cdot \vec{w}_1 = t + \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(-1 + 2|\alpha|^2 + \frac{2\text{Re}(\alpha^*\beta)}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(\frac{4\text{Re}(\alpha^*\beta)}{\sqrt{3}} \right) \quad (50)$$

$$\vec{w}_2^H \cdot H_{\vec{K}'} \cdot \vec{w}_2 = 1 + \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(-1 + 2|\eta|^2 + \frac{2\text{Re}(\eta^*\gamma)}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(\frac{4\text{Re}(\eta^*\gamma)}{\sqrt{3}} \right) \quad (51)$$

$$\vec{w}_1^H \cdot H_{\vec{K}'} \cdot \vec{w}_2 = \sqrt{3}t \left(\vec{\delta}_2 \cdot \vec{p} \right) \left(2\alpha^*\eta + \frac{\alpha^*\gamma + \beta^*\eta}{\sqrt{3}} \right) + \sqrt{3}t \left(\vec{\delta}_3 \cdot \vec{p} \right) \left(\frac{2(\alpha^*\gamma + \beta^*\eta)}{\sqrt{3}} \right) \quad (52)$$

$$\vec{w}_2^H \cdot H_{\vec{K}'} \cdot \vec{w}_1 = (\vec{w}_1^H \cdot H_{\vec{K}'} \cdot \vec{w}_2)^* \quad (53)$$

$$\Rightarrow \alpha = \frac{\sqrt{3}}{2}, \beta = \frac{1}{2}, \eta = \frac{1}{2}, \gamma = -\frac{\sqrt{3}}{2} \quad (54)$$

$$\Rightarrow \vec{w}_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \vec{w}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \quad (55)$$

These are the same basis vectors as for $\vec{K}$ just in reverse order.

$$\Rightarrow \langle H_{\vec{K}'} \rangle_{\vec{w}_1, \vec{w}_2} = t\sigma_0 + \sqrt{3}tp_y\sigma_x + \sqrt{3}tp_x\sigma_z \quad (56)$$

δH_1 and δH_2 are the same as for $\vec{K}'$ (see 39 and substitute $\vec{K} \rightarrow \vec{K}' = \vec{K}$). With the reversed order of the basis vectors this gives

$$\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2} = -t\beta \begin{pmatrix} u_{yy} & u_{yx} \\ u_{yx} & u_{xx} \end{pmatrix} \quad (57)$$

$$\langle \delta H_2 \rangle_{\vec{w}_1, \vec{w}_2} = -\frac{2\pi t}{\sqrt{3}} \begin{pmatrix} -u_{xx} & u_{yx} \\ u_{yx} & u_{xx} \end{pmatrix} = -\frac{2\pi t}{\sqrt{3}} (u_{yx}\sigma_y - u_{xx}\sigma_z) \quad (58)$$

$$\phi = \frac{t}{2}(u_{xx} + u_{yy}) \quad (59)$$

$$A_{\vec{K}'}^x = \frac{t\beta}{2\sqrt{3}}(u_{xx} - u_{yy}) + \frac{2\pi t}{3}u_{xx} \quad (60)$$

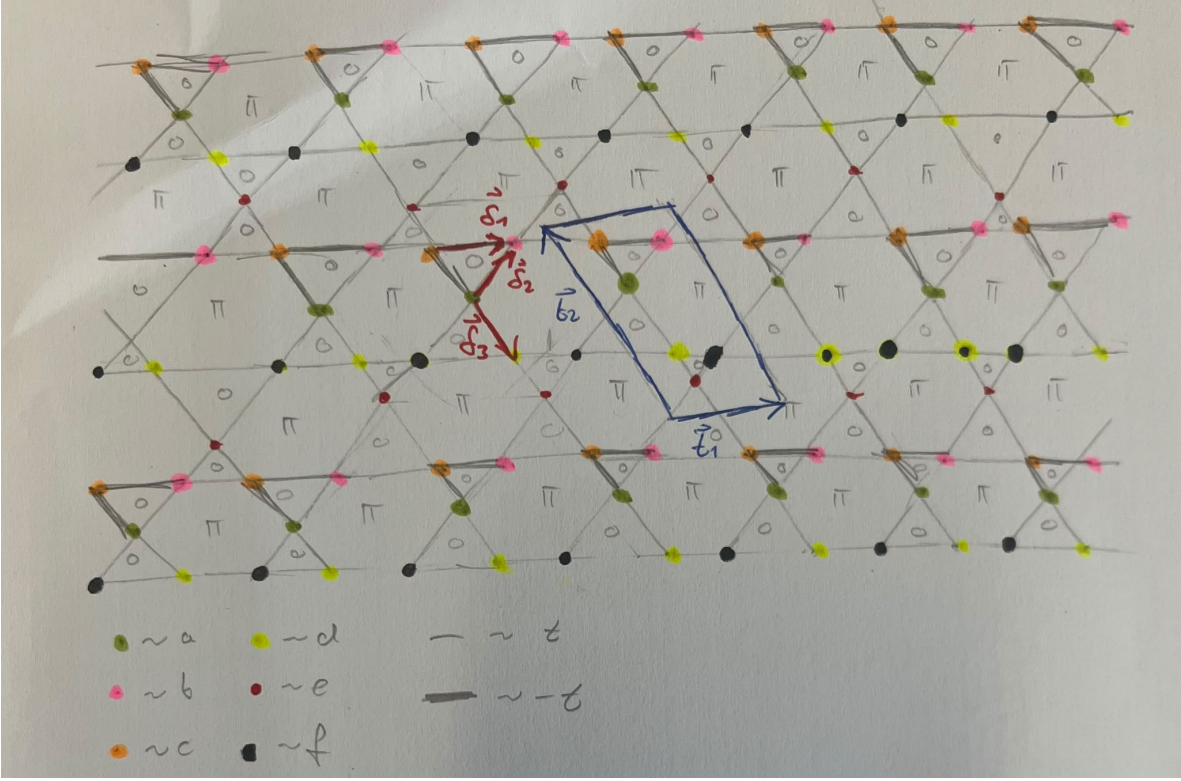
$$A_{\vec{K}'}^y = \frac{t\beta}{\sqrt{3}}u_{yx} + \frac{2\pi t}{3}u_{yx} \quad (61)$$

$$\Rightarrow A_{\vec{K}} = -A_{\vec{K}'} \quad (62)$$

0.2 π -flux State

0.2.1 Real Space

This lattice is also a triangular Bravais-lattice **that is probably not true**, only now with unit cell doubling (in the direction of $\vec{t}_2$), hence the unit cell has 6 sites. The Wigner-Seitz cell is a stretched hexagon. In the cartoon a different unit cell is shown.



The nearest neighbour vectors remain unchanged

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}. \quad (63)$$

The new primitive translations are

$$\vec{t}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = \begin{pmatrix} -2 \\ 2\sqrt{3} \end{pmatrix} \quad (64)$$

The real space (effective) tight-binding hamiltonian is

$$H = t \sum_{i\sigma} \left(-a_{i\sigma}^\dagger c_{i\sigma} - c_{i\sigma}^\dagger b_{i\sigma} + b_{i\sigma}^\dagger a_{i\sigma} + a_{i\sigma}^\dagger d_{i\sigma} + d_{i\sigma}^\dagger e_{i\sigma} + e_{i\sigma}^\dagger f_{i\sigma} + f_{i\sigma}^\dagger d_{i\sigma} \right. \\ \left. + b_{i\sigma}^\dagger c_{i+t_1;\sigma} + b_{i\sigma}^\dagger e_{i+t_1+t_2;\sigma} + f_{i\sigma}^\dagger d_{i+t_1;\sigma} + f_{i\sigma}^\dagger a_{i+t_1;\sigma} + e_{i\sigma}^\dagger c_{i-t_2;\sigma} + \text{h.c.} \right) \quad (65)$$

0.2.2 Reciprocal Space

The reciprocal lattice vectors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix} \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix} \quad (66)$$

After a Fourier transform the k -space Hamiltonian reads

$$H = t \sum_{q,\sigma} \Psi_{q\sigma}^\dagger \begin{pmatrix} 0 & e^{iq_2} & -e^{-iq_3} & 0 & e^{-iq_2} & e^{iq_3} \\ e^{-iq_2} & 0 & 2i \sin(q_1) & e^{iq_2} & 0 & 0 \\ -e^{-iq_3} & -2i \sin(q_1) & 0 & e^{-iq_3} & 0 & 0 \\ 0 & e^{-iq_2} & e^{iq_3} & 0 & e^{iq_2} & e^{-iq_3} \\ e^{iq_2} & 0 & 0 & e^{-iq_2} & 0 & 2 \cos(q_1) \\ e^{-iq_3} & 0 & 0 & e^{iq_3} & 2 \cos(q_1) & 0 \end{pmatrix} \Psi_{q\sigma}, \quad (67)$$

where

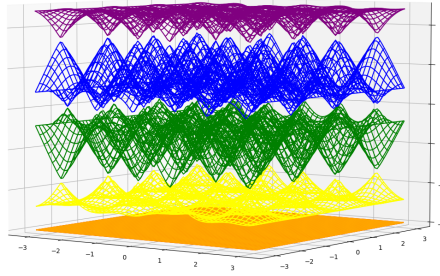
$$\Psi_{q\sigma}^\dagger = (a_{q\sigma}^\dagger \quad b_{q\sigma}^\dagger \quad c_{q\sigma}^\dagger \quad e_{q\sigma}^\dagger \quad f_{q\sigma}^\dagger \quad d_{q\sigma}^\dagger). \quad (68)$$

If I write this down somewhere serious, I should just label the sites differently.

I used the shorthand $q_i \equiv \vec{q} \cdot \vec{\delta}_i$.

0.2.3 dispersion and band trouching poiunts

The dispersion relation is shwon in the following picture

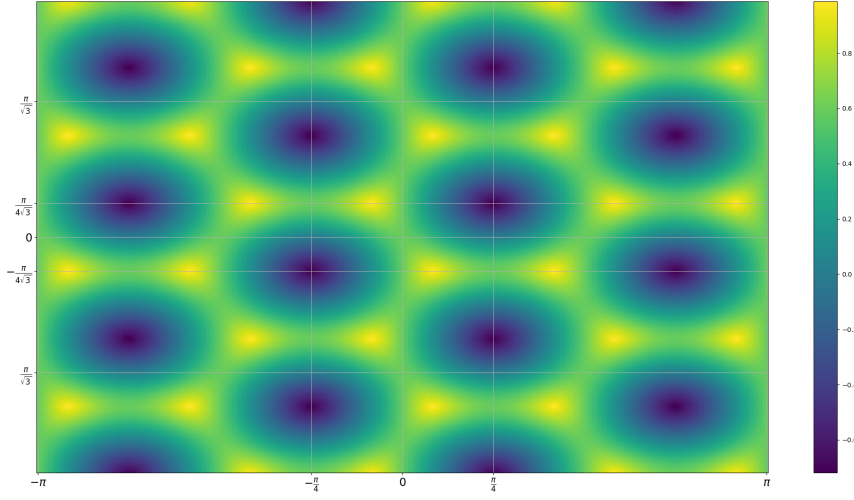


The lowest band (flat band) is doubly degenerate.

Directions on how to obtain the analytical expressions:

1. write down characteristic polynomial (probably the most annoying step)
2. Divide by the two constant roots corresponding to the flat bands (Polynomial of degree 4 remains)
3. Use symmetrie of remaining 4 bands (write as product of 2 quadratic polynomials; due to the symmetrie plane, one can guess p from the p-q-formula)
4. Solve remaining 2 quadratice equations
5. ?
6. Profit

At half filling the lowest three bands are filled, thus the Fermi-energy is between the yellow and green band.



Shown above is the 4th band (green band). Apparently there are two flavours of touching points with one of each in the first Brillouin zone. They are time reversal partners. From the numerical data these two touching points lie at

$$\vec{K} = \frac{\pi}{4} \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}, \quad \vec{K}' = -\frac{\pi}{4} \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}. \quad (69)$$

0.2.4 Linearization in reciprocal space and Projection

For now I only do the calculations for $\vec{K}$. Use

$$\vec{K} \cdot \vec{\delta}_1 = \frac{\pi}{4} \quad (70)$$

$$\vec{K} \cdot \vec{\delta}_2 = \frac{\pi}{4} \quad (71)$$

$$\vec{K} \cdot \vec{\delta}_3 = 0 \quad (72)$$

to obtain the linearized Hamiltonian in the vicinity of $\vec{K}$:

$$\begin{pmatrix} 0 & e^{i\frac{\pi}{4}}(1+ip_2) & -1+ip_3 & 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 1+ip_3 \\ e^{-i\frac{\pi}{4}}(1-ip_2) & 0 & \sqrt{2}i(1+p_1) & e^{i\frac{\pi}{4}}(1+ip_2) & 0 & 0 \\ -1-ip_3 & -\sqrt{2}i(1+p_1) & 0 & 1-ip_3 & 0 & 0 \\ 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 1+ip_3 & 0 & e^{i\frac{\pi}{4}}(1+ip_2) & 1-ip_3 \\ e^{i\frac{\pi}{4}}(1+ip_2) & 0 & 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 0 & \sqrt{2}(1-p_1) \\ 1-ip_3 & 0 & 0 & 1+ip_3 & \sqrt{2}(1-p_1) & 0 \end{pmatrix} \quad (73)$$

To keep the size reasonable maybe its a good idea to split this up into zeroth and first order in p. Might also be a good choice for storytelling purposes.

The shorthand means

$$p_i \equiv \vec{\delta}_i \cdot \vec{p},$$

also

$$e^{i\frac{\pi}{4}} = \frac{1+i}{\sqrt{2}}. \quad (74)$$

The eigenvalues for the zeroth order (/directly at the $\vec{K}$ -point) are

$$\begin{aligned} \varepsilon_1(\vec{K}) &= -2t & \varepsilon_2(\vec{K}) &= -2t & \varepsilon_3(\vec{K}) &= (1 - \sqrt{3})t \\ \varepsilon_4(\vec{K}) &= (1 - \sqrt{3})t & \varepsilon_5(\vec{K}) &= (1 + \sqrt{3})t & \varepsilon_6(\vec{K}) &= (1 + \sqrt{3})t \end{aligned} \quad (75)$$

ε_3 and ε_4 are the relevant ones.

I let mathematica calculate orthonormal eigenvectors for H at $\vec{K}$. The relevant ones (that is bands 3 and 4):

$$\vec{v}_3(\vec{K}) = \frac{1}{2\sqrt{6-3\sqrt{3}}} \left(\frac{1-\sqrt{3}}{2} + \frac{3-\sqrt{3}}{2}i, \quad \sqrt{2}(1-\sqrt{3}), \quad (-2+\sqrt{3})i, \quad \frac{1-\sqrt{3}}{2} + \frac{\sqrt{3}-3}{2}i, \quad 0, \quad 1 \right)^T \quad (76)$$

$$\vec{v}_4(\vec{K}) = \frac{1}{2\sqrt{6(7-4\sqrt{3})}} \left((3-2\sqrt{3}) + (-2+\sqrt{3})i, \quad 0, \quad i(1-\sqrt{3}), \quad 3-2\sqrt{3} + (2-\sqrt{3})i, \quad 2\sqrt{2}(2-\sqrt{3}), \quad 5-3\sqrt{3} \right)^T \quad (77)$$

The projections onto those vectors are

$$\begin{aligned} \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= (1 - \sqrt{3})t - \frac{2t}{\sqrt{3}}p_2 + \frac{(\sqrt{3}-1)t}{\sqrt{3}}p_3 \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= (1 - \sqrt{3})t + \frac{2t}{\sqrt{3}}p_2 - \frac{(\sqrt{3}-1)t}{\sqrt{3}}p_3 \\ \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= -\frac{2t}{\sqrt{3}}p_2 - \frac{(3+\sqrt{3})t}{3}p_3 \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= -\frac{2t}{\sqrt{3}}p_2 - \frac{(3+\sqrt{3})t}{3}p_3 \end{aligned} \quad (78)$$

I transform these to p_x and p_y : I only did this by hand I could easily let mathematica check this by using matrix/vector notation

$$\begin{aligned} \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= (1 - \sqrt{3})t + \frac{1-\sqrt{3}}{2}tp_x - \frac{1+\sqrt{3}}{2}tp_y \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= (1 - \sqrt{3})t - \frac{1-\sqrt{3}}{2}tp_x + \frac{1+\sqrt{3}}{2}tp_y \\ \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= -\frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= -\frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y \end{aligned} \quad (79)$$

The dispersion of this projection is

$$\begin{aligned} \varepsilon_{3\vec{K}} &= (1 - \sqrt{3})t - \sqrt{2}t\sqrt{p_x^2 + p_y^2} \\ \varepsilon_{4\vec{K}} &= (1 - \sqrt{3})t + \sqrt{2}t\sqrt{p_x^2 + p_y^2} \end{aligned} \quad (80)$$

Judging from the pictures of the dispersion I didnt expect it to be so isotropical. Check this.

I follow the usual method defining

$$\begin{aligned}\vec{w}_3 &= \alpha \vec{v}_3 + \beta \vec{v}_4 \\ \vec{w}_4 &= \eta \vec{v}_3 + \gamma \vec{v}_4\end{aligned}\tag{81}$$

Orthonormality is garuanted by

$$|\alpha|^2 + |\beta|^2 = 1\tag{82}$$

$$|\eta|^2 + |\gamma|^2 = 1\tag{83}$$

$$\alpha^* \eta + \beta^* \gamma = 0\tag{84}$$

If my counting is correct, there are 8 parameters to begin with (real and imaginary parts), the equations above eliminate 1 and 1 and 2 (the last equation are really 2 equations, one for real and one for imaginary part), leaving behind 4 free parameters. I make the following Ansatz for the leftover parameters:

$$\begin{aligned}\alpha &= \cos(\phi/2) e^{i(\chi+\theta/2)} e^{i\kappa} \\ \beta &= \sin(\phi/2) e^{i(\chi+\theta/2)} \\ \eta &= \sin(\phi/2) e^{i(\chi-\theta/2)} e^{i\kappa} \\ \gamma &= -\cos(\phi/2) e^{i(\chi-\theta/2)}\end{aligned}\tag{85}$$

If you want to be fancy about it you can think of what kind of matrix describes the relation between the two bases and then find a suitable parametrization for it. ϕ is a rotation angle, χ a global phase, θ the phase difference between the two states and κ state-internal phase. It is clear that all physical quantities can not depend on χ . At this point I might as well choose $\chi = 0$.

I find for the projection in the transformed basis:

$$\begin{aligned}\vec{w}_3^H \cdot H_{\vec{K}} \cdot \vec{w}_3 &= (1 - \sqrt{3})t \\ &+ \frac{t}{2} \left((1 - \sqrt{3}) \cos(\phi) - (1 + \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_x \\ &+ \frac{t}{2} \left(-(1 + \sqrt{3}) \cos(\phi) - (1 - \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_y\end{aligned}$$

$$\begin{aligned}\vec{w}_4^H \cdot H_{\vec{K}} \cdot \vec{w}_4 &= (1 - \sqrt{3})t \\ &+ \frac{t}{2} \left(-(1 - \sqrt{3}) \cos(\phi) + (1 + \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_x \\ &+ \frac{t}{2} \left((1 + \sqrt{3}) \cos(\phi) + (1 - \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_y\end{aligned}$$

$$\begin{aligned}\vec{w}_3^H \cdot H_{\vec{K}} \cdot \vec{w}_4 &= \frac{t}{2} \left((1 - \sqrt{3}) \sin(\phi) \cos(\theta) - (1 + \sqrt{3}) [-\cos(\phi) \cos(\theta) \cos(\kappa) + \sin(\theta) \sin(\kappa)] \right) p_x \\ &+ \frac{t}{2} \left(-(1 + \sqrt{3}) \sin(\phi) \cos(\theta) - (1 - \sqrt{3}) [-\cos(\phi) \cos(\theta) \cos(\kappa) + \sin(\theta) \sin(\kappa)] \right) p_y \\ &+ i \frac{t}{2} \left(-(1 - \sqrt{3}) \sin(\phi) \sin(\theta) - (1 + \sqrt{3}) [\cos(\theta) \sin(\kappa) + \cos(\phi) \sin(\theta) \cos(\kappa)] \right) p_x \\ &+ i \frac{t}{2} \left((1 + \sqrt{3}) \sin(\phi) \sin(\theta) - (1 - \sqrt{3}) [\cos(\theta) \sin(\kappa) + \cos(\phi) \sin(\theta) \cos(\kappa)] \right) p_y\end{aligned}$$

$$\begin{aligned}
\vec{w}_4^H \cdot H_{\vec{K}} \cdot \vec{w}_3 = & \frac{t}{2} \left((1 - \sqrt{3}) \sin(\phi) \cos(\theta) - (1 + \sqrt{3}) [-\cos(\phi) \cos(\theta) \cos(\kappa) + \sin(\theta) \sin(\kappa)] \right) p_x \\
& + \frac{t}{2} \left(-(1 + \sqrt{3}) \sin(\phi) \cos(\theta) - (1 - \sqrt{3}) [-\cos(\phi) \cos(\theta) \cos(\kappa) + \sin(\theta) \sin(\kappa)] \right) p_y \\
& + i \frac{t}{2} \left((1 - \sqrt{3}) \sin(\phi) \sin(\theta) + (1 + \sqrt{3}) [\cos(\theta) \sin(\kappa) + \cos(\phi) \sin(\theta) \cos(\kappa)] \right) p_x \\
& + i \frac{t}{2} \left(-(1 + \sqrt{3}) \sin(\phi) \sin(\theta) + (1 - \sqrt{3}) [\cos(\theta) \sin(\kappa) + \cos(\phi) \sin(\theta) \cos(\kappa)] \right) p_y
\end{aligned}$$

From these equations the $\sigma_{0,x,y,z}$ -components can be read of. [It is probably possible to simplify these equations further, considering that you dont want both momenta in the same component and the one momentum only in one component at once. For now I will just make some assumptions :\).](#)

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